

Multi-Page Visually Rich Document Understanding: A Review

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Abstract

Multi-page visually rich document understanding (MP-VRDU) has emerged as a distinct research area driven by the prevalence of long-form documents such as contracts, scientific papers, and financial reports whose evidence is distributed across pages and modalities. While single-page document understanding has matured into a well-defined sub-field with established benchmarks, the multi-page setting introduces challenges that simple extension cannot address: total document length frequently exceeds backbone context windows, answer-bearing evidence is sparsely distributed, and cross-page structural relationships cannot be recovered from per-token layout signals alone. This review synthesises recent work in MP-VRDU, organising the field by how systems manage cross-page evidence rather than by OCR dependency. We examine four architectural families, three modality-specific challenges, and the inference-time and training strategies that recur across them, before identifying three persistent gaps that current approaches have not resolved: data scarcity coupled with evaluation integrity concerns, the evidence pipeline bottleneck between coverage and fidelity, and cross-page structural reasoning that remains brittle across all current routes.

1 Introduction

Documents are the principal medium through which organisations, governments, and the scientific community record and exchange information. Contracts, financial reports, scientific papers, technical manuals, regulatory filings, and presentation decks are pervasive and structurally heterogeneous, interleaving dense text with tables, figures, diagrams, and complex layouts [1; 2]. Manually extracting information from such documents is slow, expensive, and error-prone, motivating a sustained research effort to build automated document visual question answering systems that can

answer natural-language questions over rendered document pages. For single-page inputs, this problem is largely solved. Document-specialised systems such as Qwen2.5-VL [3] and mPLUG-DocOwl1.5 [4] approach or exceed human performance on DocVQA [5], InfographicVQA [6], and ChartQA [7], and recent frontier multimodal LLMs such as Claude Opus 4.7 [8] and Gemini 3.1 Pro [9] can now read a rendered page directly from pixels, treating the document as an image and handling single-page and short multi-page inputs reliably when the input resolution is sufficient.

Real-world documents, however, almost always extend beyond a single page, and the evidence required for a realistic question is typically scattered across several pages or distributed across heterogeneous elements [1; 10; 2]. One might assume that scaling a strong long-context MLLM further would resolve this, but several structural obstacles prevent that resolution. First, input length scales with page count, with each visually rich page contributing high-resolution figures, tables, and charts that quickly exceed the context window of existing language and vision-language models; quadratic-cost self-attention [11] compounds the compute burden further, making naive scaling economically and practically infeasible at realistic document lengths. Second, the evidence required to answer a question is typically sparse and distributed across pages, requiring a system to navigate, retrieve, and reason across the entire document rather than read a single page in detail [10]; even when the document fits within a backbone’s nominal context window, long-context models attend poorly to evidence buried in the middle of long inputs [12], so simply expanding context length does not translate into proportional gains in answer quality. Third, explainability becomes more demanding, as models must not only produce an answer but also localise the pages or passages from which it was derived, a requirement that a monolithic forward pass through

a long-context backbone does not naturally support. As such, multi-page visually rich document understanding (MP-VRDU) is not a simple extension of its single-page counterpart, and together, these obstacles establish that MP-VRDU is not solved by frontier scale alone and motivates specialised architectures, evidence-selection strategies, and orchestration techniques.

This review synthesises recent MP-VRDU research, which has developed rapidly over the past two to three years and has been treated only incidentally by prior surveys that remain anchored to the single-page setting or organise progress by OCR dependency [13; 14; 15; 16; 17]. Our aim is to identify persistent gaps that motivate continued research in this area. Section 2 briefly summarises the single-page mechanisms multi-page systems inherit, orienting the discussion toward the design decisions that the multi-page setting reformulates. Section 3 introduces four architectural families that organise the field through the lens of cross-page evidence management. Sections 4 through 6 narrow into specific design dimensions: multimodal representation under context constraints, inference-time composition strategies, and component-level training. Section 7 catalogues the resources that support training and evaluation, and Section 8 identifies three persistent open gaps in the field.

2 Background: Single-Page VRDU

Multi-page VRDU systems inherit most of their per-page mechanisms from the single-page literature, recently surveyed by Ding et al. [16]. We summarise the two organising axes that recur into the multi-page setting: the OCR-dependent versus OCR-free split at the architectural level (§2.1), and three orthogonal modality-handling choices that persist independently of that split (§2.2).

2.1 Architectural Families

The dominant axis at the single-page level distinguishes two architectural families that trade different costs. **OCR-dependent frameworks** preprocess the page with off-the-shelf optical character recognition (OCR) or PDF parsers, producing recognised text and bounding-box layout that is either fed into the language model prompt directly [18; 19] or projected through an auxiliary multimodal encoder such as LayoutLMv3 [20; 21]. These systems rely on external tools to capture structural information without extensive visual pre-

training, but reliance on OCR introduces cumulative errors, particularly on handwritten or low-quality scanned documents, and the use of low-resolution document images limits the expressiveness of visual representations. **OCR-free frameworks** discard the parser and have a vision encoder read text directly from rendered page pixels [22; 23; 4], with the language model trained to recover text and layout through pretraining objectives such as text recognition, grounding, and captioning [24; 25; 26]. These frameworks enable end-to-end training and handle handwriting and complex layouts that OCR cannot, but accurate fine-grained text recognition demands high-resolution input and large-scale pretraining, which the visual compression and shape-adaptive cropping strategies discussed in §2.2 are designed to manage. The two families thus trade different costs: OCR-dependent systems inherit cumulative parser errors but avoid expensive visual pretraining, whereas OCR-free systems handle handwriting and complex layouts end-to-end but require high-resolution input and large-scale pretraining.

2.2 Multimodal Representation

Across both architectural families, three orthogonal design decisions govern how text, layout, and visual modalities are encoded, with a fourth cross-cutting choice for how they are fused. **Text** is recovered as OCR strings injected into the prompt [18; 27], through OCR-augmented multimodal encoders that produce joint text-layout-vision embeddings [21; 28], or as a training objective for the vision encoder itself through text recognition and detection tasks [24; 25; 29]. **Layout** is encoded as 2D positional features attached to text tokens [30; 28], as verbalised bounding boxes or markdown-style formatting in the prompt [19; 31; 32], as specialised layout tokens that share positional information with their associated text [33], or through supervised pretraining tasks such as visual text grounding and table reconstruction [34; 26; 35]. **Visual tokens** are managed through low-resolution patch embeddings [36; 37], high-resolution shape-adaptive cropping that divides oversized pages into fixed-size sub-images [23; 4], or compression via resamplers, Q-Formers, and convolutional reducers that reduce the visual token count by an order of magnitude [38; 39; 26]. **Modality fusion** proceeds through dedicated neural encoders such as LayoutLMv3 that produce joint multimodal embeddings [20; 21], task-oriented objectives that bind

modalities through shared supervision [4; 29], or prompt-based combinations that interleave layout descriptions with text and images at the language model input [31; 32; 21]. Single-page systems make these choices assuming the entire document fits in a single backbone input, and modality fusion operates on full page-level evidence at once. The multi-page setting reformulates each into a question of which page or region to attend to next, and modality fusion becomes a question of which evidence to combine after selective inspection rather than which to combine within a fixed input.

3 Architectural Landscape

The single-page literature is naturally organised by OCR dependency, but this axis is less informative in the multi-page setting, as OCR-using systems often treat extracted text as just one component in a multimodal pipeline [40; 41], while OCR-free systems are frequently pretrained or aligned with OCR-derived supervision [42; 4]. The defining challenge in MP-VRDU is instead how a system manages document-scale evidence: which information to retain, which to discard, and how the surviving information should interact across pages. These decisions are governed by architectural design, which determines how evidence is selected, preserved, aggregated, and exposed for reasoning. We therefore organise multi-page frameworks primarily by architectural trend, with OCR dependency as a secondary descriptor. Through this evidence-management lens, the field has shifted from encoding all pages jointly toward increasingly selective and iterative document-level handling, producing four recurring trends: page-to-document encoding models, backbone-centric adaptation, retriever-generator pipelines, and agentic pipelines.

3.1 Page-to-Document Encoding Models

Page-to-document encoding models process each page through a shared document encoder that fuses OCR text, layout, and visual features at the page level, then connect per-page representations through a document-level exchange mechanism within a single forward pass. The mechanism may take the form of per-page summary tokens [1], document-global tokens with local-global attention [10], recurrent memory states passed across pages [43], or sparse cross-page attention over long OCR sequences [2]. By preserving page identity, local layout, and reading order in the architecture

itself, this design provides a strong inductive bias for multi-page structure and is well suited to documents with regular structural templates such as multi-page forms and business reports [1; 2]. The principal limitation is intrinsic to the design: every page must still be encoded jointly in a single forward pass, computation grows with document length, and the page-to-document compression step inevitably discards fine-grained evidence as inputs scale [10; 43]. As the context windows of pretrained MLLMs have widened, the practical motivation for a dedicated cross-page encoder has weakened, and subsequent families absorb cross-page information into the backbone or select evidence before reading.

3.2 Backbone-Centric Adaptation

Backbone-centric adaptation removes the dedicated cross-page encoder entirely and processes the document inside a pretrained MLLM whose context window absorbs cross-page information directly. Adaptation strategies span a spectrum of training intensity. At the lightweight end, projectors map per-modality features into the backbone’s token space [37], adapters enable parameter-efficient fine-tuning [33], and visual-token compression modules constrain token growth on long inputs to keep multi-page documents within the context budget [38]. At the heavier end, the backbone itself is unfrozen for continued pretraining or instruction tuning on document corpora, trading parameter efficiency for stronger document-specific grounding [44], with long-context tuning further improving moderate-length multi-page performance [4; 39]. The family inherits the foundation model’s instruction-following and reasoning, avoids the engineering overhead of external retrieval or orchestration, and remains parameter-efficient to train. The cost of this design is that performance degrades beyond moderate-length documents, leaving quality bound to context length and to how faithfully fine-grained text and visual signals survive token compression [45]. Reliable text recovery depends on high-resolution input that aggressive downsampling readily compromises, and document structure such as section hierarchy and figure-caption linkage must be recovered implicitly from pixels rather than supplied through layout tokens or a structural encoder [46; 44]. Attention dilution within long contexts [12] further weakens evidence buried in the middle, motivating subsequent families that move evidence selection out of

the backbone altogether.

3.3 Retriever-Generator Pipelines

Retriever-generator pipelines decouple evidence selection from answer generation. A retrieval module returns a compact subset of OCR chunks, pages, visual patches, multimodal elements, or graph nodes, and a generator answers conditioned only on those units [47; 42]. The design adapts the retrieval-augmented generation paradigm from text [48; 49] to visually rich inputs, with the retriever instantiated as a sparse lexical scorer, a dense embedding model, or a structure-aware index, and the generator typically a pretrained MLLM. Inference cost depends on the retrieved content rather than document length, allowing the architecture to scale to documents that exceed any current MLLM context window. Modular decomposition lets retriever and generator be swapped independently, the retrieved evidence provides a partial audit trail for the answer, and the same interface extends to multi-document corpora without architectural change. Structured domains with reliable retrieval anchors such as scientific papers and technical manuals benefit most [50], and page-level visual retrieval preserves figures, tables, and layout that OCR-only pipelines lose [42; 51]. The principal limitation is the single-pass control flow, which assumes one retrieval step can surface all evidence required for the answer. Evidence missed at retrieval is unrecoverable at generation, and broadening the retrieved set introduces noise and attention dilution. Performance is further bounded by parser and embedding quality, while per-passage similarity scoring weakens when answers require aggregating weakly relevant pages or following multi-hop cross-references [52; 53], pressures that motivate replacing the fixed pipeline with adaptive, multi-step reasoning.

3.4 Agentic Pipelines

Agentic pipelines replace the fixed retrieve-then-generate control flow with iterative evidence acquisition. Drawing on the broader agentic-LLM framing [54; 55], we class as agentic any MP-VRDU system that performs multi-pass evidence acquisition or distributes work across specialised agents whose call sequence adapts at inference; multi-component systems with predetermined execution order, even when described as “agents”, remain within the retriever-generator family. One or more controller MLLMs decide what to inspect next from intermediate observations and issue tool

calls for retrieval, navigation, cropping, or OCR [56; 40]. Roles such as retrieval, inspection, synthesis, and adjudication may be distributed across agents, each backed by a different backbone selected for its subtask [57] rather than tied to a single generator. This design lifts the single-pass bottleneck, allows compute to scale with question difficulty rather than document length, and renders the trajectory itself an inspectable audit trail. The family is well suited to questions whose evidence is distributed across pages or modalities and to multi-hop questions that benefit from query reformulation between steps [41; 40], as well as visually heterogeneous documents whose text, tables, and figures benefit from tool-specific inspection [57; 58]. The flexibility carries costs. The variable-length action sequence makes inference cost difficult to bound before execution, controllers remain sensitive to prompt phrasing and may commit early to wrong evidence with silent failure, reproducibility is complicated by stochastic generation, and the pipeline inherits the recall ceiling of every retriever, parser, or OCR module it calls.

4 Multi-Page Multimodal Representation

While Section 3 categorises MP-VRDU systems by where cross-page evidence is processed, this section analyses the problem from a modality perspective. Single-page VRDU already handles text, vision, and layout through the well-established mechanisms outlined in Section 2. The multi-page setting introduces a shared pressure: the full document often exceeds the backbone’s context window, and even when it fits, the answer occupies only a small fraction of a long input where attention is diluted and key evidence is easily lost in the middle [12]. This forces each modality to trade page coverage against reading fidelity, with the resolutions differing across text, vision, and structure.

4.1 Text Modality

Text is the most information-dense modality in MP-VRDU but also the most heterogeneous in volume across pages and the most exposed to errors that compound when extracted across tens or hundreds of pages. Text reaches the backbone through two acquisition paths that recur across architectures. OCR-extracted strings support cheap indexing, natural answer-passage attribution, and easy bridging to visual elements through captions, but propagate parser errors and lose chart, layout, and

handwriting semantics; pixel-recovered text read directly by an OCR-free MLLM avoids parser dependence but demands high-resolution input and large-scale text-recognition pretraining that scale poorly with document length [22; 23]. Three patterns have emerged for handling text at document scale, each correlated with a different architectural family. **Per-page aggregation** encodes each page in detail and combines per-page representations into compact cross-page summaries through attention or memory tokens, typical of page-to-document encoders such as Hi-VT5 [1], GRAM [10], and Arctic-TILT [2]. **Whole-document compression** retains all pages within a single backbone context by aggressively reducing per-page text tokens through resamplers, layout-aware compressors, or token-reduction modules, typical of long-context backbone-centric MLLMs such as mPLUG-DocOwl2 [38] and TextHawk2 [39]. **Retrieve-then-read** abandons full-document encoding entirely, indexing text at chunk or page level and passing only a retrieved subset to the reader, characteristic of retriever-generator and agentic systems such as CREAM [59], MultiDocFusion [50], DocAgent [40], and MDocAgent [57]; agentic variants extend this into multiple selection-reading rounds. Each pattern carries a characteristic risk: per-page aggregation discards token-level detail at the summary step, whole-document compression sacrifices fine-grained reading on individual pages, and retrieve-then-read makes evidence missed at retrieval unrecoverable at the reading stage. The last two routes also depend on document-level structure to decide which pages or sections to read in detail, an issue we return to in Section 4.3.

4.2 Visual Modality

Visual content in MP-VRDU spans text-bearing structures such as dense paragraphs and tables to non-textual evidence such as charts, figures, and handwriting that OCR cannot recover. Visual encoding faces a sharper version of the coverage-fidelity tension than text, since token cost scales jointly with page count and per-page resolution while the detail required to read charts, table cells, and handwriting cannot survive aggressive downsampling. Three patterns have emerged for managing this tension. **Aggressive token compression** keeps every page available in context but reduces per-page visual tokens through resamplers or convolutional reducers, as in mPLUG-DocOwl2 [38] and TextHawk2 [39]; this preserves coverage

at the cost of per-page detail, and is often paired with OCR text to recover fine reading where the downsampled image is insufficient. **Selective high-resolution reading** restricts expensive encoding to a small subset of pages surfaced by retrieval or controller-driven inspection, sending those high-fidelity tokens into the reader while leaving other pages unread, as in Doc-V* [56] and SimpleDoc [60]. **Retrieval-only visual embeddings** decouple encoding from reading altogether: page images are encoded purely to produce retrieval embeddings, and detailed reading of selected pages is handed to a separate text or vision module, as in M3DocRAG [47] and VDocRAG [42]. The patterns are not mutually exclusive in practice; several systems pair low-resolution thumbnails for navigation with selective high-resolution inspection of pages a retriever or tool call surfaces [56; 60]. Each pattern nonetheless carries a characteristic risk: aggressive compression loses fine text and small visual elements, selective high-resolution reading inherits the recall ceiling of whichever tool selects the pages, and retrieval-only visual embeddings may rank layout-similar pages without confirming that the answer-bearing region is present.

4.3 Cross-Page Structure

Within-page mechanisms inherited from single-page VRDU, including 2D positional encodings, bounding-box coordinates, and reading-order signals attached to individual tokens, describe only per-page geometry and do not extend to relationships between sections, pages, or references. The structural problem distinctive to MP-VRDU is *cross-page* structure: section hierarchies that span multiple pages, information referenced or continued across page boundaries, and figure-to-text or table-to-caption linkages that no longer fit within a single page’s view. Two routes address it, with the second now dominant. Some backbone-centric systems train the model to predict cross-page structure end-to-end through multi-page parsing or document-level pretraining objectives [38; 44], but pretraining supervision at document scale remains scarce and expensive to obtain, limiting how far this approach can reach in practice. The dominant route exposes parsed structural information to the model directly, and three patterns recur within it. **Structure-aware chunking** uses a parser to extract section trees and heading hierarchies, then segments documents along semantic boundaries so retrieved units respect section

scope rather than sliding-window cuts, as in MultiDocFusion [50]. **Hierarchical retrieval indices** route queries first to coarse units such as sections or chapters and then to fine-grained chunks within the selected branch, reducing irrelevant pages while preserving cross-page context, as in MHier-RAG [61] and MLDocRAG [62]. **Graph-based traversal** converts headings, entities, and cross-references into a knowledge graph, allowing controllers to follow typed edges across pages rather than relying on dense similarity alone, as in KGP [52]. All three depend on parser quality: errors in heading detection, table boundary segmentation, or reading-order extraction propagate directly into every downstream retrieval and navigation step, and reliable parsers themselves remain tuned for single-page or template-bound documents rather than the heterogeneous long-form documents on which MP-VRDU is evaluated.

5 Inference-Time Adaptation

Inference-time adaptation refers to strategies that improve MP-VRDU performance through changes to the inference pipeline rather than to model parameters. A frozen language or vision-language backbone is paired with off-the-shelf retrievers, structural parsers, prompted reasoning loops, or multi-agent orchestrations, none of which require gradient updates on the backbone itself. This approach is particularly suited to MP-VRDU settings where supervised data is scarce and expensive to collect, where backbones must remain exchangeable across systems, or where rapid adaptation to new document types is needed without retraining. We organise these strategies along the inference pipeline: evidence selection through retrieval and navigation (Section 5.1), reasoning over the assembled evidence (Section 5.2), and coordination across multiple frozen components (Section 5.3).

5.1 Retrieval and Navigation

Inference-time retrieval selects a compact subset of pages, chunks, or graph nodes for the reader using off-the-shelf encoders, structural parsers, or LLM-based rerankers, without fine-tuning the retriever on document data. Retrieval quality bounds every downstream reasoning step, so the choice of signal dominates system behaviour on long documents. Two broad families have emerged, distinguished by whether passages are scored in isolation or navigated through explicit relations.

Similarity-based retrieval. Similarity-based methods score each passage against the query independently, using sparse lexical signals, dense embeddings, or combinations of both. The dominant dense signal in MP-VRDU is vision-language late-interaction encoding such as ColPali [63] or its variants, which extends the late-interaction paradigm of ColBERT [64] to page images by producing token-level query vectors and patch-level page embeddings over the rendered page, then scoring a query-page pair by taking the maximum cosine similarity of each query token over all page patches and summing the result. This MaxSim aggregation preserves fine-grained match between specific query terms and specific page regions, which is critical when a single question token may need to align with one figure caption, table cell, or paragraph buried in a long document. M3DocRAG [47] retrieves at chunk and page granularity with this signal alone, while AVIR [65] adds an adaptive selector that examines the per-query score distribution and either thresholds at a fixed cutoff for short documents or applies K-means clustering with a top- K bound on longer ones, scaling the number of returned pages with the score distribution rather than using a static K . Joint variants combine signals at the reranking stage: CREAM [59] pairs dense retrieval with iterative LLM-based reranking across overlapping chunk groups, SimpleDoc [60] combines ColPali embeddings with LLM-generated summary reranking under persistent working memory that refines the joint score across rounds, and MDocAgent [57] runs parallel text and image retrieval tracks whose outputs are aggregated by a downstream agent rather than fused at encoding time. Independent scoring handles paraphrase and cross-modal queries well, but cannot follow cross-page references or multi-hop dependencies that no single passage carries.

Structure-based retrieval. Structure-based methods exploit a parsed representation of the document to make passage relations first-class objects. Tree-based variants such as MultiDocFusion [50] and DocAgent [40] extract section hierarchies and outlines and group passages along semantic boundaries so retrieved chunks preserve parent context, while MHier-RAG [61] extends this by routing queries through dual in-page and cross-page indices with LLM-based reranking across both. Graph-based variants treat headings,

entities, and cross-references as graph nodes with typed edges, allowing controllers to traverse from one passage to related ones rather than relying on similarity alone. KGP [52] constructs a knowledge graph over passages and document-structure nodes such as pages, paragraphs, and tables with edges encoding lexical similarity and structural belonging, then uses an instruction-tuned LLM as a traverser that predicts the next evidence node from the visited path. MoLoRAG [53] builds a page graph in which edge candidates are filtered by embedding similarity and then scored for logical relevance by a frozen vision-language prompt, combining semantic and logical signals for multi-hop traversal. MLDocRAG [62] constructs the graph per query, anchoring chunk nodes to LVLM-generated query nodes so the structure adapts to the question rather than being fixed at index time. Structure-based methods recover references that similarity scoring misses, at the cost of construction overhead, dependence on parser quality, and traversal cost that grows with hop count.

5.2 Inference-Time Reasoning Strategies

Inference-time reasoning strategies govern how a frozen language or vision-language model accesses, updates, and reasons over evidence during decoding, without modifying model parameters. In MP-VRDU, these strategies can be broadly categorised by whether evidence acquisition remains static or becomes dynamically interactive during inference. **Closed-buffer reasoning** operates over a fixed evidence set assembled before generation begins, whereas **tool-interactive reasoning** allows the model to iteratively acquire additional evidence through retrieval or page-level actions during the reasoning process.

Closed-buffer reasoning. Closed-buffer reasoning performs multi-step reasoning over a fixed evidence buffer assembled by a prior retrieval step or by a compressed document representation. Chain-of-thought prompting [66] is the canonical instantiation, eliciting intermediate reasoning steps that integrate textual, tabular, and visual evidence before producing the final answer. MHier-RAG [61] applies structured chain-of-thought prompts over hierarchical retrieval outputs, although a single reasoning chain remains vulnerable to compounding errors on dense layouts. SimpleDoc [60] adds self-refinement and self-verification prompts that emit

revised queries when the assembled evidence is judged insufficient, but the loop terminates on the backbone’s own judgement and inherits its blind spots. Closed-buffer reasoning is fundamentally bounded by retrieval completeness: evidence absent from the buffer cannot be recovered by subsequent reasoning or verification, regardless of how many self-checks are applied.

Tool-interactive reasoning. Tool-interactive reasoning interleaves reasoning and evidence acquisition during inference, allowing newly observed pages to guide subsequent reasoning steps. The ReAct paradigm [67] is the canonical instantiation, alternating intermediate reasoning traces with tool calls whose outputs feed the next reasoning stage. Doc-React [68] instantiates this with flat retrieval, InfoNCE-guided sub-query refinement, and entropy-based stopping, although each iteration still selects from a single retriever’s ranked list and recovery from a poor encoder match relies on subsequent query rewrites. Outline-driven variants combine ReAct with the structured navigation of Section 5.1: DocAgent [40] traverses a tree-formatted XML outline through section-level search and fetch tools chosen turn-by-turn by an actor agent, inheriting the parser’s outline quality and degrading on documents without exploitable structure. A complementary refinement is coarse-to-fine visual reading, in which fetched pages are inspected at multiple resolutions rather than at a fixed scale so that high-resolution encoding is reserved for regions surfaced by the coarse pass, concentrating compute on task-relevant regions but making reasoning conditional on the localiser since the fine pass cannot recover when the coarse pass misroutes. Adaptive evidence acquisition improves localisation and supports more exploratory multi-page reasoning, but introduces tool latency that grows with rounds, sensitivity to prompt design and to early action choices that subsequent steps cannot recover from, and attention dilution within fetched pages at their reading resolution.

5.3 Agentic Orchestration

Retriever-generator and agentic pipelines often distribute a single MP-VRDU question across multiple specialised language or vision-language model calls, each handling a particular modality, reasoning step, or verification role. The frozen backbones are not modified; coordination is achieved through prompt design and call ordering. The two fami-

lies correspond directly to whether the trajectory of calls is fixed at design time or adapts to the question at inference: retriever-generator pipelines run a trajectory fixed in advance regardless of how many agents they invoke, while agentic pipelines treat the call sequence itself as a variable.

Fixed-trajectory pipelines. Fixed-trajectory pipelines follow a control flow set in advance: which calls are made, in what order, and how their outputs are combined are all determined at design time. Modality-partitioned variants dispatch readers by input type and aggregate their outputs through a final synthesis step. MDocAgent [57] runs five parallel agents in which a general agent provides initial multimodal context, a critical agent extracts key evidence, dedicated text and image agents read their respective tracks, and a summariser fuses the four outputs into a single answer. M2RAG [69] splits work between a text filter, a visual extractor, and a modal fuser that applies a confidence-conditional rule, falling back to one modality when the other reports no information and fusing only when both contribute. Such pipelines improve coverage relative to single-agent baselines, but lack an error-correction stage: a wrong choice of routing or ordering cannot be revised mid-execution, and an incorrect output from one branch can dominate aggregation when the other branch lacks counter-evidence.

Adaptive-trajectory pipelines. Adaptive-trajectory pipelines treat the call sequence itself as a variable: which agents are invoked and how many rounds the loop runs are decided per question rather than fixed in advance. Loop-based variants such as ViDoRAG [41] cycle a seeker that retrieves coarse candidates, an inspector that reviews them at fine resolution and emits reflective feedback, and an answerer that performs a consistency check, with a Gaussian mixture model setting per-query bounds from the similarity distribution rather than using a static top- K . Proposer-verifier variants insert an explicit verification stage between candidate generation and final answer: DocLens [58] samples multiple candidates and routes them through an adjudicator that compares reasoning paths and selects the most consistent conclusion rather than fusing them unconditionally, while DocAgent [40] pairs an actor with a reviewer that cross-checks through complementary tools and writes disagreements into a shared reflection memory that updates the actor’s behaviour on

subsequent rounds. MACT [70] decomposes the loop further into planning, execution, judgement, and answer agents, separating the judge from the corrector to avoid the same agent excusing its own errors. Adaptive-trajectory pipelines improve the accuracy-latency trade-off, but inference cost remains hard to bound before execution and the verifier typically draws on the same backbone family as the proposer, so shared blind spots can persist when both agents fail in the same way.

6 Training Strategies

The inference-time strategies of Section 5 hold model parameters fixed, which keeps backbones exchangeable and avoids expensive supervision but leaves system behaviour bound to what off-the-shelf encoders, prompts, and orchestration can achieve. Many MP-VRDU systems therefore extend these pipelines by training specific components, replacing brittle retrieval signals with learned ones, prompted reasoning with internalised reasoning policies, and prompt-only orchestration with distilled or reinforcement-learned trajectories. Training entry points mirror the inference-time decomposition, with the retriever, the reasoner, and the agentic orchestration policy each amenable to targeted optimisation while the rest of the pipeline remains frozen. We organise this section accordingly, addressing each of the three components in turn, followed by two cross-cutting techniques, parameter-efficient tuning and multi-stage curriculum training, that recur across all three targets.

6.1 Trained Retrieval Components

Off-the-shelf encoders such as ColPali [63] score pages in isolation, leaving three gaps that retriever training aims to address. Modality-alignment variants pretrain a visual document encoder against OCR text: VDocRAG [42] introduces representation-compression retrieval and representation-compression generation objectives that compress page images into single-vector embeddings while preserving alignment with their OCR-derived textual content, retaining text alignment without the storage and latency cost of late-interaction patch embeddings. Cross-page-fusion variants freeze the encoder and learn a neighbour-blending head: DFVC [71] adds a multi-layer perceptron adapter with gating and residual connections that fuses each page embedding with its neighbours, so evidence spread across adjacent pages,

which independent per-page scoring would split, becomes retrievable as a unit. Logical-relevance variants train a small scorer over a page graph: MoLoRAG [53] fine-tunes a frozen Qwen2.5-VL scorer to rank neighbour pages during graph traversal, supplying the cross-page logical signal that embedding similarity alone often misses and effectively turning retrieval into a learned multi-hop walk rather than a single-step similarity match.

6.2 Trained Reasoning Components

Prompted reasoning leaves quality sensitive to prompt design and base-model behaviour, motivating approaches that train the reasoning component directly. Reward-driven variants train a reasoning policy with explicit evidence supervision: DocR1 [72] fine-tunes a vision-language backbone with Evidence-Page-Guided GRPO, whose composite reward combines format compliance, answer accuracy, and a page-citation indicator that explicitly credits trajectories which cite the correct evidence pages, producing structured outputs that interleave thinking traces, evidence pages, and answers. Trace-imitation variants instil a reasoning chain through supervised fine-tuning and preference optimisation: Chain-of-Reading [73] bakes in a four-stage plan-execute-integrate-verify chain through a staged pipeline of LoRA supervised tuning on a mixed corpus, full-parameter tuning on a curated reasoning dataset, and direct preference optimisation on preference pairs that prefer faithful chains over hallucinated shortcuts. Uncertainty-driven variants train an aggregation head rather than a trajectory: DocSLM [46] learns an entropy-based calibrator that selects the lowest-uncertainty answer across streaming page segments, providing a self-consistency signal that scales to long documents under tight parameter budgets without requiring full-document context.

6.3 Trained Agentic Orchestration

Prompt-only orchestration is unreliable on smaller open-weight backbones that cannot follow ReAct protocols. Behavioural-cloning variants distil agent trajectories from a stronger model: DocDancer [74] fine-tunes a Qwen3 agent on synthesised exploration-then-synthesis traces that pair `Search` and `Read` ReAct actions with intent-conditioned tool selection, training the smaller model to match the trajectory shape of a larger teacher rather than learning the underlying policy from scratch. Reinforcement-learned variants re-

ward correct tool use and trajectory shape: Doc-V* [56] couples supervised fine-tuning on coarse-to-fine ReAct traces with GRPO over a two-action space of `retrieval_page` and `fetch_page` calls plus entropy-based stopping that learns when to commit to an answer rather than fetch another page, while MACT [70] trains a multi-agent policy with agent-wise adaptive test-time scaling that allocates sample budgets across the planner, executor, judge, and answerer roles per question, spending more samples on the role that the question’s difficulty most stresses rather than running every agent at a fixed budget.

6.4 Cross-Cutting Training Techniques

Two techniques recur across retriever, reasoner, and orchestrator training, independent of which component is being optimised.

Parameter-Efficient Tuning. Full backbone fine-tuning is prohibitive at long-document context lengths and erases pretrained capabilities, so the dominant pattern freezes the backbone and trains lightweight modules through parameter-efficient tuning approaches such as LoRA [75], projection heads, or task-specific adapters. LayTokenLLM [33] freezes the language model and adds a layout tokenisation module plus LoRA, encoding spatial layout through compressed positional tokens with negligible context overhead. CREAM [59] applies the LLaMA-Adapter-V2 recipe above a frozen Pix2Struct encoder, training only LoRA matrices, attention pooling, and bias and normalisation parameters in its answerer. The same pattern appears in DFVC’s frozen-ColPali adapter and DocDancer’s LoRA trajectory tuning, confirming that the choice between parameter-efficient and full fine-tuning is more often a compute-budget decision than a target-component one.

Multi-Stage Curriculum Training. Several systems converge on multi-stage curricula that start on single-page or local objectives before introducing document-scale supervision, exploiting the fact that text reading and layout understanding transfer from single-page data while cross-page integration must be learned separately. mPLUG-DocOwl2 [38] uses three stages of single-image structure learning on DocStruct4M, MP-DocStruct1M continued pretraining, and multi-task fine-tuning. DocSLM [46] extends this to a five-stage schedule that adds image-OCR alignment at the start and streaming-abstention calibration at the end, demonstrating

that the curriculum can be lengthened to cover both pre-compression alignment and post-reasoning uncertainty calibration without abandoning the staged structure.

7 Datasets

Datasets in MP-VRDU span three roles: pretraining, fine-tuning, and benchmarking. Pretraining historically relied on single-page archives such as IIT-CDIP [76] and OCR-IDL [77], with genuinely multi-page corpora only recently emerging through DocStruct4M’s mixed single- and multi-page pretraining [4] and the purely multi-page MP-DocStruct1M, Doc-750K, and scientific PaperPDF [38; 44; 51]. Fine-tuning has shifted toward bespoke synthetic generation per model, with strong vision-language models writing question-answer or trajectory data over scraped PDFs since per-page annotation does not scale to long documents [37; 60; 72; 74]. Benchmarks have evolved from mid-length extractive evaluation on MP-DocVQA, DUDE, and SlideVQA [1; 78; 79] to long-context multimodal evaluation on MMLongBench-Doc, LongDocURL, and DocBench [80; 81; 82], with retrieval and page-localisation increasingly scored alongside answer accuracy [83; 72; 58]. Five benchmarks (MP-DocVQA, MMLongBench-Doc, DUDE, DocVQA, and InfographicVQA [6]) appear in roughly half or more of recent MP-VRDU papers and constitute the de-facto axes for cross-model comparison. A detailed catalogue with per-dataset statistics is provided in Table ??.

8 Open Gaps

The preceding sections have organised MP-VRDU through four architectural families, three modality-specific challenges, the inference-time compositions and training recipes that recur across them, and the dataset ecosystem that supports them. Read together, these strands surface three persistent gaps that cut across the architectural taxonomy and remain unresolved by any current approach, motivating continued research in this area.

8.1 Data Scarcity and Evaluation Integrity

The first gap concerns the data foundation on which MP-VRDU progress is measured. Per-page annotation does not scale to documents averaging eight to twenty pages, leaving most systems to rely on vision-language-model-synthesised question-answer pairs and trajectories generated

over scraped PDFs [37; 60; 72; 74]. This pragmatic response to scarcity has produced a structural integrity problem. Of the twelve papers in this review that introduce a synthesised corpus, nine source their PDFs from existing evaluation benchmarks, and MP-DocVQA alone appears in 15 fine-tuning mixes and 17 evaluation suites among the 36 surveyed papers, with DUDE and DocVQA showing similar overlap. Performance reported as zero-shot therefore routinely includes the test corpus in its training history, with models trained on a strong vision-language model’s behaviour over benchmark PDFs and then evaluated against those same PDFs under metrics often judged by the same model family. Compounding this, no paper in the surveyed corpus reports synthetic-data quality audits, document-hash de-duplication against test pools, or backbone-level contamination checks, while agentic stochasticity and per-paper benchmark curations block direct cross-system comparison even when contamination is absent. Dominant benchmarks remain overwhelmingly English-centric, leaving non-English layouts, scripts, and document conventions underrepresented. The cumulative effect is that headline numbers in the surveyed literature cannot be taken as faithful indicators of progress, and the field lacks the evaluation infrastructure required to distinguish genuine capability gains from contamination or synthetic-evaluation artefacts.

8.2 The Evidence Pipeline Bottleneck

The second gap concerns a constraint that no architectural family has overcome. MP-VRDU systems must reconcile high-resolution per-page reading with document-scale evidence aggregation, an objective that exceeds any MLLM’s compute and context budget [80]. Every architectural family takes a lossy compromise, but the compromise sits at a different stage of the pipeline. Page-to-document encoders lose fine-grained evidence at the per-page compression step, since summary or memory tokens cannot retain every token-level detail. Backbone-centric MLLMs dilute attention across long contexts and lose evidence buried in the middle of the input [12]. Retriever-generator pipelines drop everything outside the top- K and depend on the assumption that relevance is rankable by query similarity in a single step. Agentic systems inherit the recall ceiling of every tool they call, so weaknesses in any retriever or parser bound the entire trajectory regardless of how many rounds

the controller runs. Compression is also typically fixed at index time and independent of the question, so evidence is discarded before the question is seen, and retrieval-stage chunking errors, OCR artefacts, parser failures, and per-passage scoring that ignores cross-references propagate directly into the answer. Iterative retrieval lifts this ceiling partially yet inherits the recall of every tool it calls. The bottleneck is fundamental rather than incidental: across page-to-document, backbone-centric, retriever-generator, and agentic families, no architecture has decoupled answer quality from a coverage-versus-fidelity trade-off; current architectures merely shift where the trade-off is made.

8.3 Cross-Page Structural Reasoning

The third gap concerns a problem distinctive to the multi-page setting that single-page mechanisms cannot address. Within-page layout is largely inherited from single-page VRDU through 2D positional embeddings and spatial pretraining (Section 2), while cross-page structure such as section hierarchies that span multiple pages, page order, multi-page tables, footnote-to-text linkages, and figure-caption-text references spread across pages remains the distinctive multi-page problem and has no analogous inherited solution. Three current routes each carry a distinct cost that prevents adoption at scale. Layout-as-objective approaches train the backbone to predict cross-page structure end-to-end through multi-page parsing or document-level pretraining objectives [38; 44], but demand supervision that is expensive at document scale, since reliable cross-page structural annotation is difficult to collect at the volumes pretraining requires. Implicit pixel-based recovery depends on backbone capability and high-resolution input that aggressive downsampling readily compromises, so the same compression pressures discussed in Section 4.2 weaken the structural signal whenever pages are compressed to fit document-scale inputs. Explicit parsed structure inherits parser fragility on headings, table boundaries, and reading order, with errors propagating into every downstream retrieval and navigation step; the parsers themselves remain tuned for single-page or template-bound documents rather than the heterogeneous long-form documents on which MP-VRDU is evaluated, and rarely recover relationships such as theorem-proof or footnote-reference linkages that cross page boundaries. The result is that cross-page structural reasoning currently sits between three imperfect routes, none of

which addresses the problem end-to-end at the cost and reliability that production MP-VRDU systems would require.

9 Future Research Directions

The three gaps in Section 8 point toward several concrete research directions that the MP-VRDU community can pursue.

Addressing data scarcity and evaluation integrity requires investment in evaluation infrastructure that has so far lagged behind model development. Held-out closed-test benchmarks that are not used in any pretraining or fine-tuning corpus would establish baseline cross-system comparability, while mandatory contamination declarations covering pretraining, fine-tuning, and backbone leakage would allow the field to distinguish genuine generalisation from artefacts of overlapping corpora. Generator-evaluator separation, in which the model that synthesises training data is structurally prevented from also judging evaluation outputs, would break the current circular evaluation regime. Trajectory-level metrics paired with retrieval and grounding scores, building on the evidence-aware evaluation seen in MMLongBench-Doc [80], LongDocURL [81], and recent reasoning-augmented systems [72; 58], would surface the localisation failures that answer-only accuracy masks. Finally, multilingual corpora covering non-English layouts, scripts, and document conventions would extend the field’s reach beyond its current English-centric scope.

Addressing the evidence pipeline bottleneck requires rethinking compression and retrieval as question-dependent rather than question-independent operations. Query-conditional and modality-aware compression that retains evidence based on the question’s evidential demands, rather than discarding fine-grained content at index time, would push the trade-off later in the pipeline where it can adapt to the query. Retrieval that follows cross-references and logical relations rather than scoring passages in isolation would address the multi-hop and aggregation questions that single-pass similarity retrieval currently fails on, building on the early graph-traversal and hierarchical-index work surveyed in Section 5.1 but extending it toward retrievers that can plan over document structure rather than reacting to it. Hybrid architectures that combine selective high-resolution reading with structure-aware navigation, rather than picking one compression strategy at design time, may offer a

path toward decoupling answer quality from the coverage-fidelity trade-off rather than merely shifting it.

Addressing cross-page structural reasoning requires advances on both the supervision and representation fronts. Cheaper structural supervision, perhaps through weakly supervised approaches that leverage publication metadata, table-of-contents signals, or document-rendering pipelines, could mitigate the annotation cost that currently limits layout-as-objective training. Learned cross-page representations that survive compression would address the implicit-recovery route’s dependence on high-resolution input, possibly by training compression modules with explicit structural reconstruction objectives. End-to-end structure-aware retrieval that does not depend on offline parser output would close the loop on the parser-fragility problem, treating structure inference as a learned component of the retrieval pipeline rather than a brittle preprocessing step. Across all three directions, progress will depend on benchmarks that explicitly evaluate cross-page structural reasoning rather than treating it as an incidental dimension of answer accuracy.

Beyond these three threads, the MP-VRDU community would benefit from broader coordination on terminology, evaluation protocols, and reporting standards. The rapid succession of architectural families documented in this review has produced inconsistent vocabulary even for closely related ideas, and cross-family comparisons remain difficult to draw without shared evaluation protocols. As the field matures past the current period of rapid architectural innovation, consolidation around shared infrastructure may prove as valuable as further model development.

10 LLM Usage and Other Notes

LLM Usage. Large language models, specifically Claude Opus 4.7 and GPT 5.5, were used throughout the preparation of this review for spelling and grammar correction, sentence-level rewording, reading-flow refinement, and other writing-quality tasks. All conceptual content, argumentation, and selection of cited works are my own. The LLM was used as a writing aid rather than as a source of claims, and all factual statements, citations, and quantitative figures have been verified against primary sources before inclusion. Intellectual responsibility for the final document rests with the author.

Relationship to a Companion Survey Paper.

Sections 3–8 were adapted from a survey paper on the MP-VRDU frontier of which I am the first author. The corresponding material was largely written by me prior to any co-author contributions to the survey. The adaptation in this review trims, re-tones, and reorients the survey material toward the methodology research direction motivated in §§8–??.

Supporting Materials. The following supporting artefacts were produced during the preparation of this review:

- Paper inventory and cross-reference spreadsheet (surveyed systems, dataset and benchmark coverage, evaluation metrics):
<https://docs.google.com/spreadsheets/d/1ZoPeIvnZKkHOQN8G3Fjj9gY8jnLImZvcYQrL36nj/edit?usp=sharing>
- Project repository (literature review L^AT_EX source, build script, bibliography, and supporting code):
<https://github.com/LeweiXu/MP-VRDU/>
- Overleaf project for the companion survey paper: *URL pending*.

References

- [1] R. Tito, D. Karatzas, and E. Valveny, “Hierarchical multimodal transformers for multipage docvqa,” *Pattern Recognition*, vol. 144, p. 109834, 2023. [Online]. Available: <https://doi.org/10.1016/j.patcog.2023.109834>
- [2] L. Borchmann, M. Pietruszka, W. Jaśkowski, D. Jurkiewicz, P. Halama, P. Józiak, L. Garncarek, P. Liskowski, K. Szyndler, A. Gretkowski *et al.*, “Arctic-tilt. business document understanding at sub-billion scale,” in *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 6: Industry Track)*, 2025, pp. 264–283.
- [3] S. Bai, K. Chen, X. Liu, J. Wang, W. Ge, S. Song, K. Dang, P. Wang, S. Wang, J. Tang *et al.*, “Qwen2.5-vl technical report (no. arxiv: 2502.13923). arxiv,” 2025.
- [4] A. Hu, H. Xu, J. Ye, M. Yan, L. Zhang, B. Zhang, J. Zhang, Q. Jin, F. Huang, and J. Zhou, “mplug-docowl 1.5: Unified structure learning for ocr-free document understanding,” in *Findings of the Association for Computational Linguistics: EMNLP*

- 2024, 2024, pp. 3096–3120. [Online]. Available: <https://aclanthology.org/2024.findings-emnlp.175>
- [5] M. Mathew, D. Karatzas, and C. Jawahar, “Docvqa: A dataset for vqa on document images,” in *Proceedings of the IEEE/CVF winter conference on applications of computer vision*. IEEE, 2021, pp. 2200–2209. [Online]. Available: <https://doi.org/10.1109/WACV48630.2021.00225>
 - [6] M. Mathew, V. Bagal, R. Tito, D. Karatzas, E. Valveny, and C. Jawahar, “Infographicvqa,” in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 2022, pp. 1697–1706.
 - [7] A. Masry, X. L. Do, J. Q. Tan, S. Joty, and E. Hoque, “Chartqa: A benchmark for question answering about charts with visual and logical reasoning,” in *Findings of the association for computational linguistics: ACL 2022*, 2022, pp. 2263–2279.
 - [8] Anthropic, “Claude Opus 4.7 system card,” <https://www.anthropic.com/claude-opus-4-7-system-card>, Apr. 2026, accessed: 2026-05-12.
 - [9] Google DeepMind, “Gemini 3.1 Pro model card,” <https://storage.googleapis.com/deepmind-media/Model-Cards/Gemini-3-1-Pro-Model-Card.pdf>, Feb. 2026, accessed: 2026-05-12.
 - [10] T. Blau, S. Fogel, R. Ronen, A. Golts, R. Ganz, E. Ben Avraham, A. Aberdam, S. Tsiper, and R. Litman, “Gram: Global reasoning for multi-page vqa,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 15 598–15 607.
 - [11] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, “Attention is all you need,” *Advances in neural information processing systems*, vol. 30, 2017.
 - [12] N. F. Liu, K. Lin, J. Hewitt, A. Paranjape, M. Bevilacqua, F. Petroni, and P. Liang, “Lost in the middle: How language models use long contexts,” *Transactions of the association for computational linguistics*, vol. 12, pp. 157–173, 2024.
 - [13] N. Subramani, A. Matton, M. Greaves, and A. Lam, “A survey of deep learning approaches for ocr and document understanding,” *arXiv preprint arXiv:2011.13534*, 2020.
 - [14] Y. Ding, S. C. Han, J. Lee, and E. Hovy, “Deep learning based visually rich document content understanding: A survey,” *Artificial Intelligence Review*, 2026.
 - [15] H. Gbada, K. Kalti, and M. A. Mahjoub, “Deep learning approaches for information extraction from visually rich documents: datasets, challenges and methods,” *International Journal on Document Analysis and Recognition*, vol. 28, no. 1, pp. 121–142, 2025.
 - [16] Y. Ding, S. Luo, Y. Dai, Y. Jiang, Z. Li, G. Martin, and Y. Peng, “A survey on mllm-based visually rich document understanding: Methods, challenges, and emerging trends,” *arXiv preprint arXiv:2507.09861*, 2025.
 - [17] S. Gao, S. Zhao, X. Jiang, L. Duan, Y. X. Chng, Q.-G. Chen, W. Luo, K. Zhang, J.-W. Bian, and M. Gong, “Scaling beyond context: A survey of multimodal retrieval-augmented generation for document understanding,” *arXiv preprint arXiv:2510.15253*, 2025.
 - [18] D. Wang, N. Raman, M. Sibue, Z. Ma, P. Babkin, S. Kaur, Y. Pei, A. Nourbakhsh, and X. Liu, “Docllm: A layout-aware generative language model for multimodal document understanding,” in *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Association for Computational Linguistics, 2024, pp. 8529–8548. [Online]. Available: <https://doi.org/10.18653/v1/2024.acl-long.463>
 - [19] J. He, L. Wang, Y. Hu, N. Liu, H. Liu, X. Xu, and H. T. Shen, “Icl-d3ie: In-context learning with diverse demonstrations updating for document information extraction,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision*. IEEE, 2023, pp. 19 485–19 494. [Online]. Available: <https://doi.org/10.1109/ICCV51070.2023.01785>
 - [20] Y. Huang, T. Lv, L. Cui, Y. Lu, and F. Wei, “Layoutlmv3: Pre-training for document ai with unified text and image masking,” in *Proceedings of the 30th ACM International Conference on Multimedia*. ACM, 2022, pp. 4083–4091. [Online]. Available: <https://doi.org/10.1145/3503161.3548112>
 - [21] C. Luo, Y. Shen, Z. Zhu, Q. Zheng, Z. Yu, and C. Yao, “Layoutllm: Layout instruction tuning with large language models for document understanding,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 2024, pp. 15 630–15 640. [Online]. Available: <https://doi.org/10.1109/CVPR52733.2024.01480>
 - [22] G. Kim, T. Hong, M. Yim, J. Park, J. Yim, W. Hwang, S. Yun, D. Han, and S. Park, “Donut: Document understanding transformer without ocr,” *arXiv preprint arXiv:2111.15664*, vol. 7, no. 15, p. 2, 2021.
 - [23] J. Ye, A. Hu, H. Xu, Q. Ye, M. Yan, G. Xu, C. Li, J. Tian, Q. Qian, J. Zhang et al., “Ureader: Universal ocr-free visually-situated language understanding with multimodal large language model,” in *Findings of the Association for Computational Linguistics: EMNLP 2023*, 2023, pp. 2841–2858. [Online]. Available: <https://doi.org/10.18653/v1/2023.findings-emnlp.187>
 - [24] Y. Wang, W. Zhou, H. Feng, K. Zhou, and H. Li, “Towards improving document understanding: An exploration on text-grounding via mllms,”

- arXiv preprint arXiv:2311.13194*, 2023. [Online]. Available: <https://arxiv.org/abs/2311.13194>
- [25] H. Feng, Z. Wang, J. Tang, J. Lu, W. Zhou, H. Li, and C. Huang, “Unidoc: A universal large multi-modal model for simultaneous text detection, recognition, spotting and understanding,” *arXiv preprint arXiv:2308.11592*, 2023.
- [26] Y. Liu, B. Yang, Q. Liu, Z. Li, Z. Ma, S. Zhang, and X. Bai, “Textmonkey: An ocr-free large multimodal model for understanding document,” 2024. [Online]. Available: <https://arxiv.org/abs/2403.04473>
- [27] S. Kim, H. Liao, S. Appalaraju, P. Tang, Z. Tu, R. K. Satzoda, R. Manmatha, V. Mahadevan, and S. Soatto, “Dockd: Knowledge distillation from llms for open-world document understanding models,” in *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Association for Computational Linguistics, 2024, pp. 3167–3193. [Online]. Available: <https://aclanthology.org/2024.emnlp-main.185>
- [28] Y. Zhu, Y. Zhang, D. Liu, C. Xie, Z. Xiong, B. Zheng, and S. Guo, “Enhancing document understanding with group position embedding: A novel approach to incorporate layout information,” in *The Thirteenth International Conference on Learning Representations*, 2025. [Online]. Available: <https://openreview.net/forum?id=Dj9a4zQsSl>
- [29] Z. Wang, T. Guan, P. Fu, C. Duan, Q. Jiang, Z. Guo, S. Guo, J. Luo, W. Shen, and X. Yang, “Marten: Visual question answering with mask generation for multi-modal document understanding,” 2025. [Online]. Available: <https://arxiv.org/abs/2503.14140>
- [30] Y. Xu, M. Li, L. Cui, S. Huang, F. Wei, and M. Zhou, “Layoutlm: Pre-training of text and layout for document image understanding,” in *Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*. ACM, 2020, pp. 1192–1200. [Online]. Available: <https://doi.org/10.1145/3394486.3403172>
- [31] M. Lamott, Y.-N. Weweler, A. Ulges, F. Shafait, D. Krechel, and D. Obradovic, “Lapdoc: Layout-aware prompting for documents,” in *International Conference on Document Analysis and Recognition*. Springer, 2024, pp. 142–159. [Online]. Available: https://doi.org/10.1007/978-3-031-70546-5_9
- [32] V. Perot, K. Kang, F. Luisier, G. Su, X. Sun, R. S. Boppana, Z. Wang, Z. Wang, J. Mu, H. Zhang *et al.*, “Lmdx: Language model-based document information extraction and localization,” in *Findings of the Association for Computational Linguistics: ACL 2024*, 2024, pp. 15 140–15 168. [Online]. Available: <https://doi.org/10.18653/v1/2024.findings-acl.899>
- [33] Z. Zhu, C. Luo, Z. Shao, F. Gao, H. Xing, Q. Zheng, and J. Zhang, “A simple yet effective layout token in large language models for document understanding,” *arXiv preprint arXiv:2503.18434*, 2025. [Online]. Available: <https://arxiv.org/abs/2503.18434>
- [34] C. Liu, K. Yin, H. Cao, X. Jiang, X. Li, Y. Liu, D. Jiang, X. Sun, and L. Xu, “Hrvda: High-resolution visual document assistant,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 15 534–15 545. [Online]. Available: <https://doi.org/10.1109/CVPR52733.2024.01471>
- [35] W. Liao, J. Wang, H. Li, C. Wang, J. Huang, and L. Jin, “Doclayllm: An efficient and effective multi-modal extension of large language models for text-rich document understanding,” *arXiv preprint arXiv:2408.15045*, 2024. [Online]. Available: <https://doi.org/10.48550/arXiv.2408.15045>
- [36] X. Xie, H. Yan, L. Yin, Y. Liu, J. Ding, M. Liao, Y. Liu, W. Chen, and X. Bai, “Pdf-wukong: A large multimodal model for efficient long pdf reading with end-to-end sparse sampling,” 2024. [Online]. Available: <https://arxiv.org/abs/2410.05970>
- [37] R. Tanaka, T. Iki, K. Nishida, K. Saito, and J. Suzuki, “Instructdoc: A dataset for zero-shot generalization of visual document understanding with instructions,” in *Proceedings of the AAAI conference on artificial intelligence*. AAAI Press, 2024, pp. 19 071–19 079. [Online]. Available: <https://doi.org/10.1609/aaai.v38i17.29874>
- [38] A. Hu, H. Xu, L. Zhang, J. Ye, M. Yan, J. Zhang, Q. Jin, F. Huang, and J. Zhou, “mplug-docowl2: High-resolution compressing for ocr-free multi-page document understanding,” in *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2025, pp. 5817–5834.
- [39] Y.-Q. Yu, M. Liao, J. Zhang, and J. Wu, “Texthawk2: A large vision-language model excels in bilingual ocr and grounding with 16x fewer tokens,” 2024. [Online]. Available: <https://arxiv.org/abs/2410.05261>
- [40] L. Sun, L. He, S. Jia, Y. He, and C. You, “Docagent: An agentic framework for multi-modal long-context document understanding,” in *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, 2025, pp. 17 712–17 727.
- [41] Q. Wang, R. Ding, Z. Chen, W. Wu, S. Wang, P. Xie, and F. Zhao, “Vidorag: Visual document retrieval-augmented generation via dynamic iterative reasoning agents,” 2025. [Online]. Available: <https://arxiv.org/abs/2502.18017>
- [42] R. Tanaka, T. Iki, T. Hasegawa, K. Nishida, K. Saito, and J. Suzuki, “Vdocrag: Retrieval-augmented generation over visually-rich documents,” in *Proceedings of the Computer Vision and Pattern Recognition Conference*, 2025, pp. 24 827–24 837.
- [43] Q. Dong, L. Kang, and D. Karatzas, “Multi-page document vqa with recurrent memory transformer,” in *International Workshop on Document Analysis Systems*. Springer, 2024, pp. 57–70.

- [44] Y. Duan, Z. Chen, Y. Hu, W. Wang, S. Ye, B. Shi, L. Lu, Q. Hou, T. Lu, H. Li *et al.*, “Docopilot: Improving multimodal models for document-level understanding,” in *Proceedings of the Computer Vision and Pattern Recognition Conference*, 2025, pp. 4026–4037.
- [45] M. Jia, W. Yu, K. Ma, T. Fang, Z. Zhang, S. Ouyang, H. Zhang, M. Jiang, and D. Yu, “Leopard: A vision language model for text-rich multi-image tasks,” *arXiv preprint arXiv:2410.01744*, 2024. [Online]. Available: <https://doi.org/10.48550/arXiv.2410.01744>
- [46] T. Hannan, D. Mallios, P. Pathak, F. Sardari, T. Seidl, G. Bertasius, M. Fayyaz, and S. Sengupta, “Docslm: A small vision-language model for long multimodal document understanding,” *arXiv preprint arXiv:2511.11313*, 2025.
- [47] J. Cho, D. Mahata, O. Irsoy, Y. He, and M. Bansal, “M3docrag: Multi-modal retrieval is what you need for multi-page multi-document understanding,” *arXiv preprint arXiv:2411.04952*, 2024.
- [48] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W.-t. Yih, T. Rocktäschel, S. Riedel, and D. Kiela, “Retrieval-augmented generation for knowledge-intensive NLP tasks,” in *Advances in Neural Information Processing Systems*, vol. 33, 2020, pp. 9459–9474.
- [49] K. Guu, K. Lee, Z. Tung, P. Pasupat, and M.-W. Chang, “REALM: Retrieval-augmented language model pre-training,” in *Proceedings of the 37th International Conference on Machine Learning*. PMLR, 2020, pp. 3929–3938.
- [50] J. Shin, C. Park, J. Park, J. Seo, and H.-S. Lim, “Multidocfusion: Hierarchical and multimodal chunking pipeline for enhanced rag on long industrial documents,” in *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, 2025, pp. 20 996–21 015.
- [51] X. Xie, H. Yan, L. Yin, Y. Liu, J. Ding, M. Liao, Y. Liu, W. Chen, and X. Bai, “Wukong: A large multimodal model for efficient long pdf reading with end-to-end sparse sampling,” *arXiv preprint arXiv:2410.05970*, 2024.
- [52] Y. Wang, N. Lipka, R. A. Rossi, A. Siu, R. Zhang, and T. Derr, “Knowledge graph prompting for multi-document question answering,” in *Proceedings of the AAAI conference on artificial intelligence*, vol. 38, no. 17, 2024, pp. 19 206–19 214.
- [53] X. Wu, Y. Tan, N. Hou, R. Zhang, and H. Cheng, “Molorag: Bootstrapping document understanding via multi-modal logic-aware retrieval,” in *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, 2025, pp. 14 035–14 056.
- [54] A. Plaat, M. van Duijn, N. Van Stein, M. Preuss, P. van der Putten, and K. J. Batenburg, “Agentic large language models, a survey,” *Journal of Artificial Intelligence Research*, vol. 84, 2025.
- [55] L. Wang, C. Ma, X. Feng, Z. Zhang, H. Yang, J. Zhang, Z. Chen, J. Tang, X. Chen, Y. Lin *et al.*, “A survey on large language model based autonomous agents,” *Frontiers of Computer Science*, vol. 18, no. 6, p. 186345, 2024.
- [56] Y. Zheng, P. Fu, H. Li, Z. Wang, Y. Zhang, W. Ruan, X. Zhang, Z. Wei, Z. Luo, J. Luan *et al.*, “Doc-v*: Coarse-to-fine interactive visual reasoning for multi-page document vqa,” *arXiv preprint arXiv:2604.13731*, 2026.
- [57] S. Han, P. Xia, R. Zhang, T. Sun, Y. Li, H. Zhu, and H. Yao, “Mdocagent: A multi-modal multi-agent framework for document understanding,” 2025. [Online]. Available: <https://arxiv.org/abs/2503.13964>
- [58] D. Zhu, R. Meng, J. Chen, S. Li, T. Pfister, and J. Yoon, “Doclens: A tool-augmented multi-agent framework for long visual document understanding,” *arXiv preprint arXiv:2511.11552*, 2025.
- [59] J. Zhang, Y. Yu, and Y. Zhang, “Cream: coarse-to-fine retrieval and multi-modal efficient tuning for document vqa,” in *Proceedings of the 32nd ACM International Conference on Multimedia*, 2024, pp. 925–934.
- [60] C. Jain, Y. Wu, Y. Zeng, J. Liu, S. Dai, Z. Shao, Q. Wu, and H. Wang, “SimpleDoc: Multi-Modal document understanding with Dual-Cue page retrieval and iterative refinement,” in *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, C. Christodoulopoulos, T. Chakraborty, C. Rose, and V. Peng, Eds. Suzhou, China: Association for Computational Linguistics, November 2025, pp. 28 398–28 415. [Online]. Available: <https://aclanthology.org/2025.emnlp-main.1443/>
- [61] Z. Gong, C. Mai, and Y. Huang, “Mhier-rag: Multi-modal rag for visual-rich document question-answering via hierarchical and multi-granularity reasoning,” *arXiv preprint arXiv:2508.00579*, 2025.
- [62] Y. Zhang and Y. Wu, “Mldocrag: Multimodal long-context document retrieval augmented generation,” *arXiv preprint arXiv:2602.10271*, 2026.
- [63] M. Faysse, H. Sibille, T. Wu, B. Omrani, G. Viaud, C. Hudelot, and P. Colombo, “Colpali: Efficient document retrieval with vision language models,” *arXiv preprint arXiv:2407.01449*, 2024.
- [64] O. Khattab and M. Zaharia, “ColBERT: Efficient and effective passage search via contextualized late interaction over BERT,” in *Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2020, pp. 39–48.

- [65] Z. Li, Y. Li, L. Kang, D. Karatzas, and W. Ma, “Avir: Adaptive visual in-document retrieval for efficient multi-page document question answering,” in *Proceedings of the 7th ACM International Conference on Multimedia in Asia*, 2025, pp. 1–7.
- [66] J. Wei, X. Wang, D. Schuurmans, M. Bosma, F. Xia, E. Chi, Q. V. Le, D. Zhou *et al.*, “Chain-of-thought prompting elicits reasoning in large language models,” *Advances in neural information processing systems*, vol. 35, pp. 24 824–24 837, 2022.
- [67] S. Yao, J. Zhao, D. Yu, N. Du, I. Shafran, K. Narasimhan, and Y. Cao, “React: Synergizing reasoning and acting in language models,” *arXiv preprint arXiv:2210.03629*, 2022.
- [68] J. Wu, Y. Xia, T. Yu, X. Chen, S. S. Harsha, A. V. Maharaj, R. Zhang, V. Bursztyn, S. Kim, R. A. Rossi *et al.*, “Doc-react: Multi-page heterogeneous document question-answering,” in *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, 2025, pp. 67–78.
- [69] T. Duan, M. Hu, L. Xue, C. Liu, G. Geng, W. Luo, and Z. Luo, “M2rag: A multi-agent and multimodal fusion framework for retrieval-augmented document qa,” in *International Conference on Neural Information Processing*. Springer, 2025, pp. 507–520.
- [70] X. Yu, C. Xu, Z. Chen, Y. Zhang, S. Lu, C. Yang, J. Zhang, S. Yan, and X. Hu, “Visual document understanding and reasoning: A multi-agent collaboration framework with agent-wise adaptive test-time scaling,” *arXiv preprint arXiv:2508.03404*, 2025.
- [71] B. Qian, K. Deng, Y. Wu, J. Zhang, J. Sun, Y. Xue, and C. Fan, “Accurate multi-page document retrieval by effectively fusing context information across pages,” *Electronics*, vol. 15, no. 7, p. 1353, 2026.
- [72] J. Xiong, Y. Wang, W. Zhao, C. Liu, B. Yin, W. Zhou, and H. Li, “Docrl: Evidence page-guided grpo for multi-page document understanding,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 40, no. 13, 2026, pp. 11 178–11 186.
- [73] J. Guo, C. Huang, H. Tao, Z. An, G. Ke, and X. Fang, “End-to-end document understanding via chain-of-reading,” 2026.
- [74] Q. Zhang, X. Lv, J. Wu, B. Li, Z. Tao, G. Yan, H. Zhang, B. Wang, J. Xu, H. Mi *et al.*, “Docdancer: Towards agentic document-grounded information seeking,” *arXiv preprint arXiv:2601.05163*, 2026.
- [75] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, “LoRA: Low-rank adaptation of large language models,” in *International Conference on Learning Representations*, 2022.
- [76] D. Lewis, G. Agam, S. Argamon, O. Frieder, D. Grossman, and J. Heard, “Building a test collection for complex document information processing,” in *Proceedings of the 29th annual international ACM SIGIR conference on Research and development in information retrieval*. ACM, 2006, pp. 665–666. [Online]. Available: <https://doi.org/10.1145/1148170.1148307>
- [77] A. F. Biten, R. Tito, L. Gomez, E. Valveny, and D. Karatzas, “OCR-IDL: OCR annotations for industry document library dataset,” in *European Conference on Computer Vision Workshops (Text in Everything)*, 2022.
- [78] J. Van Landeghem, R. Tito, L. Borchmann, M. Pietruszka, P. Joziak, R. Powalski, D. Jurkiewicz, M. Coustaty, B. Anckaert, E. Valveny *et al.*, “Document understanding dataset and evaluation (dude),” in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023, pp. 19 528–19 540.
- [79] R. Tanaka, K. Nishida, K. Nishida, T. Hasegawa, I. Saito, and K. Saito, “Slidevqa: A dataset for document visual question answering on multiple images,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 37, no. 11, 2023, pp. 13 636–13 645.
- [80] Y. Ma, Y. Zang, L. Chen, M. Chen, Y. Jiao, X. Li, X. Lu, Z. Liu, Y. Ma, X. Dong *et al.*, “Mmlongbench-doc: Benchmarking long-context document understanding with visualizations,” *Advances in Neural Information Processing Systems*, vol. 37, pp. 95 963–96 010, 2024.
- [81] C. Deng, J. Yuan, P. Bu, P. Wang, Z.-Z. Li, J. Xu, X.-H. Li, Y. Gao, J. Song, B. Zheng, and C.-L. Liu, “LongDocURL: a comprehensive multimodal long document benchmark integrating understanding, reasoning, and locating,” in *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, W. Che, J. Nabende, E. Shutova, and M. T. Pilehvar, Eds. Vienna, Austria: Association for Computational Linguistics, July 2025, pp. 1135–1159. [Online]. Available: <https://aclanthology.org/2025.acl-long.57/>
- [82] A. Zou, W. Yu, H. Zhang, K. Ma, D. Cai, Z. Zhang, H. Zhao, and D. Yu, “DOCBENCH: A benchmark for evaluating LLM-based document reading systems,” *arXiv preprint arXiv:2407.10701*, 2024.
- [83] Y. Hui, Y. Lu, and H. Zhang, “Uda: A benchmark suite for retrieval augmented generation in real-world document analysis,” *Advances in Neural Information Processing Systems*, vol. 37, pp. 67 200–67 217, 2024.